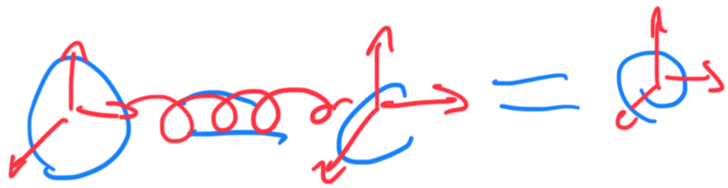
CHEMISTRY

Lewis Structure



They can move freely but only together (bonds)

$3(N)$ dimension is 3 times Atom

$= 9 = \sigma$

$\vec{X} = f(\sigma)$

$F = -k(\vec{X})$

$\downarrow$

$U = \frac{1}{2} k (\Delta x)$

Translation

Atom wise

DoF

3 atom → 3 Degree of Freedom

possibilities of Movement

$O = C = O$

$O = C = O$

$O = C = O$

Center of

Co-ord atoms Dimension molecule

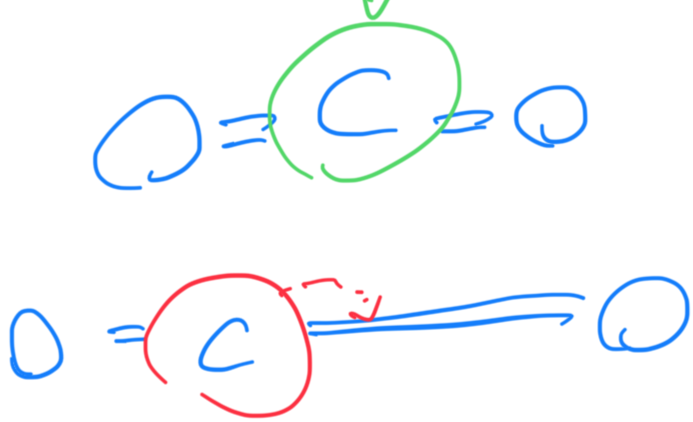
3 > 3 > 9 < 1 <

Mass

x, y, z

Rotation

Vibration

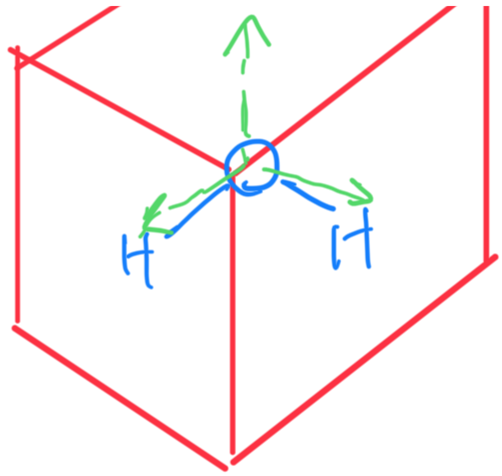


Some weird case

it pulls to middle

So position can be

<u>estimate</u>



(We Use)

Alignment

Eliminate these

bc those aren't in

Potential Function

VIBRATION

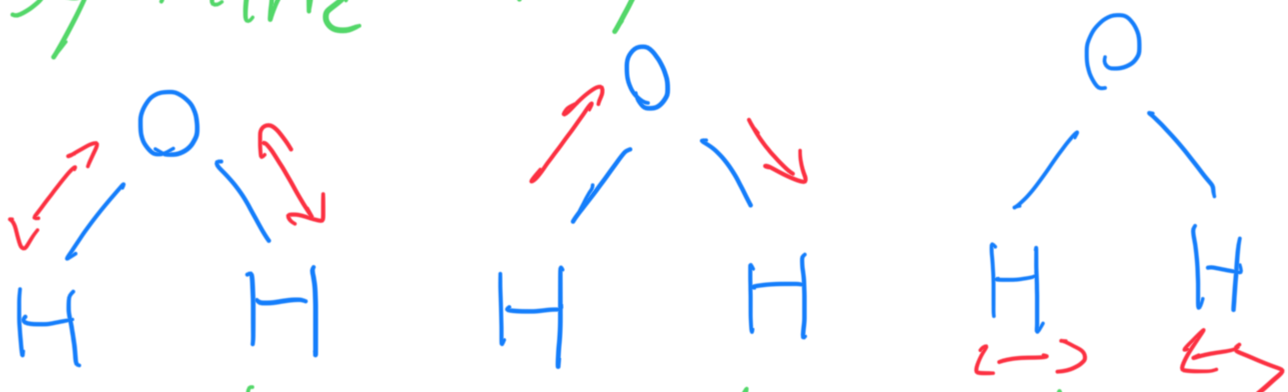
modes

How atom is place

With respect to

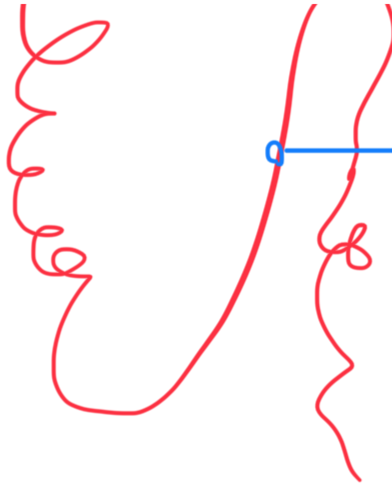
other atoms

Symmetric Asymmetric Wagging



stretch strech Strech

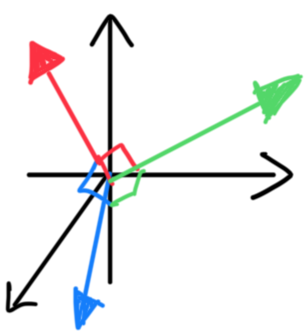
The Protein we're working has 46 AA



estimated size of H_2O in comparison

orthogonal Google: 90°

ตั้งฉาก



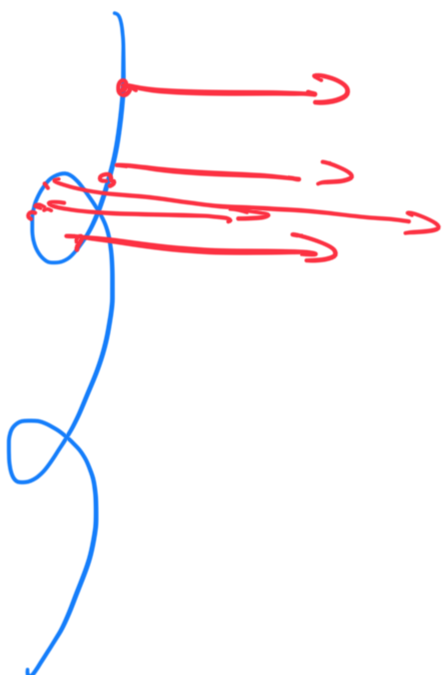
Joseph: you need them both

x, y, z

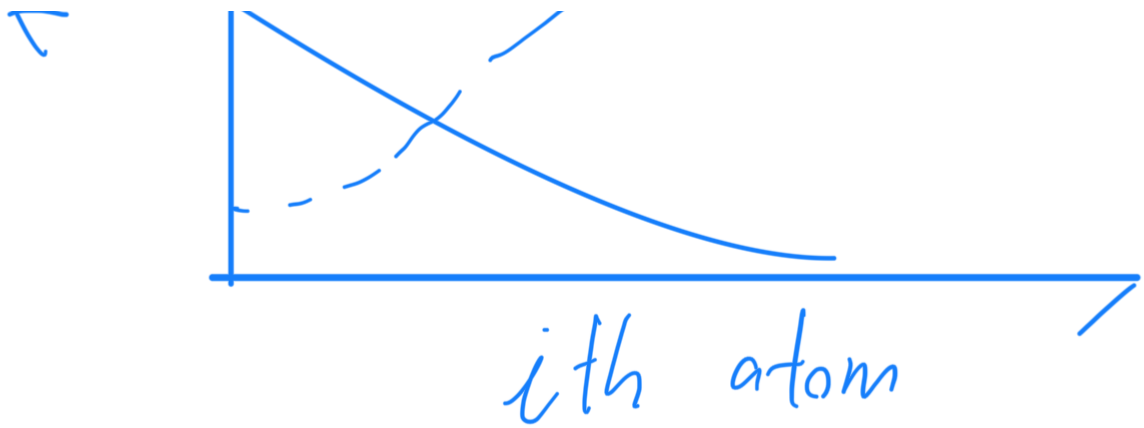
☆ Visualization Job!

How does each atom move w.r.t this

Latents



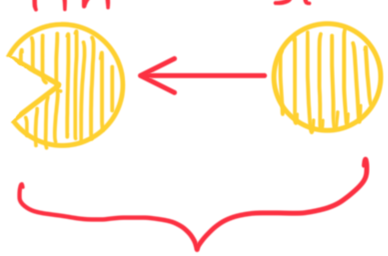
RMSFluctuation



RMSD

difference in time

fin start



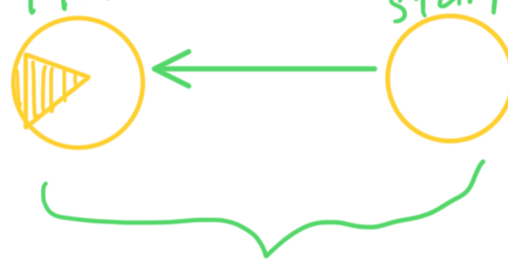
Large Change through RMSD

RMSF

individual residue flexibility

How Much a particular residue moves (fluctuates) during a simulation

fin start



Only Moved Atoms have High RMSF

during a simulatio

= whole time?

average result?